The Effect of Paid Search Advertising on Revenue: A Replication of the eBay Natural Experiment

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1 Introduction

The research question was trying to find the causal effectiveness of paid search ads on eBay's revenue. This matters because while firms spend billions on SEM, measuring its causal effect is difficult. A natural experiment is used to answer the question, where eBay turned off ads in some markets.

2 Experimental Setting

Blake et al. (2014) convinced eBay to stop bidding on AdWords. They did this by doing the following. Firstly, this was not a fully randomized trial – the largest DMAs were excluded from treatment, so treatment and control groups differ in baseline revenue. Therefore, the paper uses differences-in-difference (DID) rather than simple comparison of means. 65 of 210 DMAs were selected for treatment starting May 22, 2012. Treatment lasted 8 weeks, with the treatment group having their search turned off, and the control group having their search kept on.

3 Data and Figures

The following is daily revenue for 210 DMAs from April to July 2012. The variables include date, DMA identifier, treatment period, group, and revenue.

While both groups show similar weekly oscillation patterns, it is the control group has higher average revenue (the largest DMAs have been excluded from treatment). Additionally, it is also hard to see a clear divergence visually at this scale.

The above figure shows the log ratio of control to treatment revenue. The line fluctuates around a stable level before May 22 and appearing to increase slightly after. This visual pattern motivates the formal DID test.

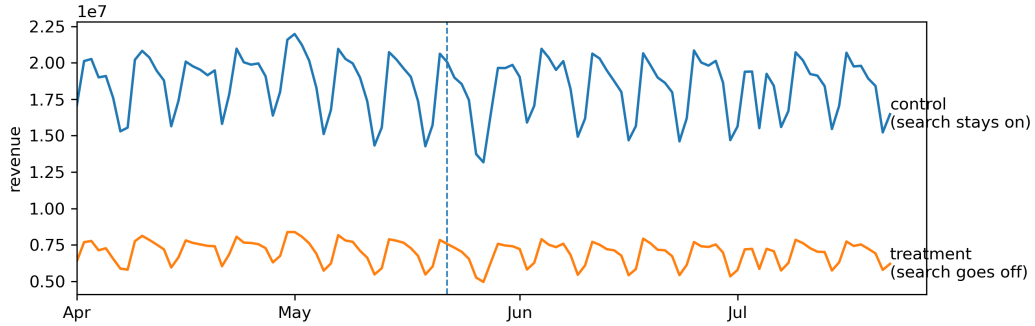

Figure 1: Average revenue for treatment and control DMAs over time. The dashed line marks May 22, 2012, when SEM was turned off for the treatment group.

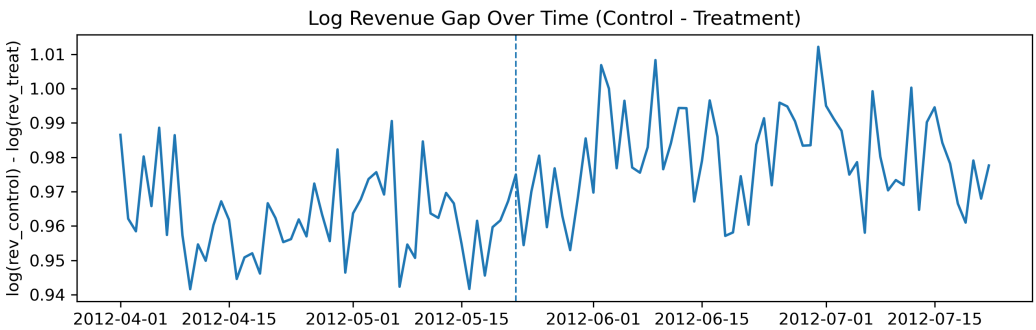

Figure 2: Log-scale average revenue difference between control and treatment groups over time. The gap appears to increase after May 22.

4 Results

The point estimate $\hat{\gamma} = -0.0066$ in log scale. In levels, this means it equals $\exp(-0.0066) = 0.9934$, showing treatment DMAs lost about 0.66 percent of revenue when paid search was turned off. The 95 percent confidence interval is $[-0.0175, 0.0043]$, including zero, implying the effect is not statistically significant at the 5 percent level. The economic interpretation would be that turning paid search off had a very small (and statistically insignificant) negative effect on revenue. This suggests that for a well-known brand like eBay, most users would find the site through organic search anyway.

5 Conclusion

The DID analysis finds a small, statistically insignificant effect. This replicates the main finding of Blake et al. (2014). Limitations of this replication include the following: the data is scaled/shifted (not real revenue figures); the treatment was not fully randomized; we use a simple DID rather than regression with controls. Overall, the result has important implications for marketing ROI measurement.

Table 1: Difference-in-Differences Estimate of the Effect of Paid Search on Revenue

	Log Scale	Level Scale
Point Estimate ($\hat{\gamma}$)	-0.0066	0.9934
Standard Error	0.0056	—
95% CI	$[-0.0175,\ 0.0043]$	$[0.9827,\ 1.0043]$

References

[1] Blake, T., C. Nosko, and S. Tadelis (2014). "Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment." *Econometrica*, 83(1): 155–174.

[2] Taddy, M. (2019).

Business Data Science. McGraw-Hill, Chapter 5.